

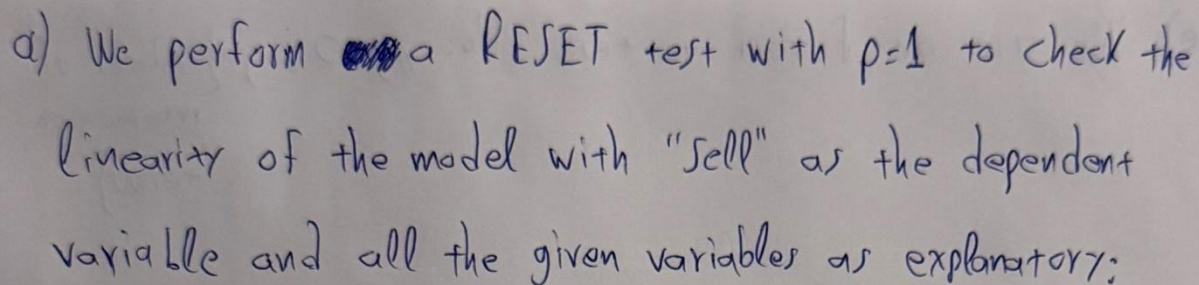
Econometrics: Methods and Applications

Case Project

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GitHub repo: [BillDimitro/econometrics-methods-and-applications](https://github.com/BillDimitro/econometrics-methods-and-applications)

Note: This PDF presents the final numerical results and interpretation for the Case Project. All regressions, statistical tests and numerical computations were implemented in Python. The corresponding source code for this exercise is available in Module_8/scripts/case_project.py, which is included in the GitHub repository referenced above. The code is extensively documented with comments explaining the methodology, implementation, and interpretation of the results.



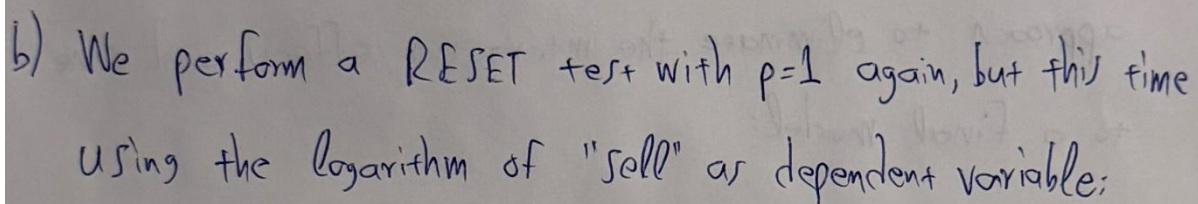
a) We perform ~~the~~ a RESET test with $p=1$ to check the linearity of the model with "sell" as the dependent variable and all the given variables as explanatory:

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Using p = 1, the RESET test for the model with sell as dependent variable yields an F-statistic = 26.986, with a p-value = 0.0.
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Since p < 0.05, we reject the null of correct linear specification at the 5% level. The linear model with the sale price in levels appears to be misspecified.
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This suggests that the relationship between house price and the explanatory variables is not adequately captured by a simple linear model in the original sale price.
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b) We perform a RESET test with $p=1$ again, but this time using the logarithm of "sell" as dependent variable:

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Using p = 1, the RESET test for the model with log(sell) as dependent variable yields an F-statistic = 0.27, with a p-value = 0.603.
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Since p > 0.05, we fail to reject the null of correct linear specification at the 5% level. Therefore, using log(sell) as the dependent variable improves the functional form of the model.
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c) We estimate a model that includes both the "lot" variable and its logarithm, as well as all other explanatory variables without transformation. From now on, we use $\log(\text{sell})$ as dependent variable.

Regressing $\log(\text{sell})$ on a constant, $\log(\text{lot})$, and all other variables yields:

Cons ->	b = 7.15	SE = 0.683	t-statistic = 10.469	p-value = 0.0
Lot ->	b = -0.0	SE = 0.0	t-statistic = -0.918	p-value = 0.359
LogLot ->	b = 0.383	SE = 0.091	t-statistic = 4.219	p-value = 0.0
Bdms ->	b = 0.035	SE = 0.014	t-statistic = 2.442	p-value = 0.015
Fb ->	b = 0.166	SE = 0.02	t-statistic = 8.161	p-value = 0.0
Sty ->	b = 0.091	SE = 0.013	t-statistic = 7.224	p-value = 0.0
Drv ->	b = 0.107	SE = 0.028	t-statistic = 3.752	p-value = 0.0
Rec ->	b = 0.055	SE = 0.026	t-statistic = 2.078	p-value = 0.038
Ffin ->	b = 0.105	SE = 0.022	t-statistic = 4.848	p-value = 0.0
Ghw ->	b = 0.179	SE = 0.044	t-statistic = 4.079	p-value = 0.0
Ca ->	b = 0.164	SE = 0.021	t-statistic = 7.657	p-value = 0.0
Gar ->	b = 0.048	SE = 0.011	t-statistic = 4.203	p-value = 0.0
Reg ->	b = 0.134	SE = 0.023	t-statistic = 5.884	p-value = 0.0

R-squared = 0.687

When both lot size and log lot size are included in the same regression, lot size itself is not statistically significant $p = 0.359 > 0.05$, while $\log(\text{lot})$ is strongly statistically significant. Therefore, the evidence supports using the logarithm of lot size rather than lot size in levels.

d) We consider a model with $\log(\text{sell})$ as dependent variable, and as explanatory factors the $\log(\text{lot})$, all other variables as before, and 10 interaction terms, each formed by multiplying $\log(\text{lot})$ with one of the other explanatory variables. Below lies the estimation of this model.

Main effects:

Cons ->	b = 8.966	SE = 1.071	t-statistic = 8.375	p-value = 0.0
LogLot ->	b = 0.153	SE = 0.128	t-statistic = 1.19	p-value = 0.235
Bdms ->	b = 0.019	SE = 0.327	t-statistic = 0.058	p-value = 0.953
Fb ->	b = -0.368	SE = 0.429	t-statistic = -0.858	p-value = 0.391
Sty ->	b = 0.489	SE = 0.31	t-statistic = 1.579	p-value = 0.115
Drv ->	b = -1.463	SE = 0.717	t-statistic = -2.04	p-value = 0.042
Rec ->	b = 1.674	SE = 0.656	t-statistic = 2.552	p-value = 0.011
Ffin ->	b = -0.032	SE = 0.446	t-statistic = -0.071	p-value = 0.943
Ghw ->	b = -0.506	SE = 0.903	t-statistic = -0.56	p-value = 0.575
Ca ->	b = -0.34	SE = 0.496	t-statistic = -0.686	p-value = 0.493
Gar ->	b = 0.402	SE = 0.259	t-statistic = 1.554	p-value = 0.121
Reg ->	b = 0.118	SE = 0.48	t-statistic = 0.247	p-value = 0.805

Interaction effects:

Loglot_Bdms ->	b = 0.002	SE = 0.039	t-statistic = 0.054	p-value = 0.957
Loglot_Fb ->	b = 0.062	SE = 0.05	t-statistic = 1.237	p-value = 0.217
Loglot_Sty ->	b = -0.046	SE = 0.036	t-statistic = -1.29	p-value = 0.198
Loglot_Drv ->	b = 0.192	SE = 0.087	t-statistic = 2.193	p-value = 0.029
Loglot_Rec ->	b = -0.188	SE = 0.076	t-statistic = -2.468	p-value = 0.014
Loglot_Ffin ->	b = 0.016	SE = 0.053	t-statistic = 0.301	p-value = 0.763
Loglot_Ghw ->	b = 0.081	SE = 0.107	t-statistic = 0.759	p-value = 0.448
Loglot_Ca ->	b = 0.06	SE = 0.058	t-statistic = 1.026	p-value = 0.305
Loglot_Gar ->	b = -0.041	SE = 0.03	t-statistic = -1.372	p-value = 0.171
Loglot_Reg ->	b = 0.002	SE = 0.056	t-statistic = 0.027	p-value = 0.978

R-squared = 0.695

Among the 10 interaction effects between log lot size and the other explanatory variables, 2 are individually significant at the 5% level: the interaction with driveway and the interaction with recreational room.

e) We use an F-test to test $H_0: \gamma_j = 0$, $j = \{\text{indices of interaction terms}\}$

The F-test for the joint significance of the interaction terms yields:

F-statistic = 1.471, with a p-value = 0.147.

Therefore, we fail to reject the null hypothesis that $\gamma_j = 0$, where j are the interaction terms indices.

Although 2 interaction effects were individually significant in question (d), the interaction effects are not jointly significant at the 5% level.

Therefore, as a group, the interactions between log lot size and the other explanatory variables do not add statistically significant explanatory power to the model.

f) We follow the general-to-specific model specification approach to eliminate the interaction effects and come to a final model:

Regression 1:
Removing log_lot_reg (p-value = 0.978).

Regression 2:
Removing log_lot_bdms (p-value = 0.958).

Regression 3:
Removing log_lot_ffin (p-value = 0.743).

Regression 4:
Removing log_lot_ghw (p-value = 0.437).

Regression 5:
Removing log_lot_ca (p-value = 0.302).

Regression 6:
Removing log_lot_gar (p-value = 0.245).

Regression 7:
Removing log_lot_fb (p-value = 0.187).

Regression 8:
Removing log_lot_sty (p-value = 0.291).

Regression 9:
Removing log_lot_drv (p-value = 0.050).

Final model reached after 9 variable eliminations.

Remaining variables: ['const', 'log_lot', 'bdms', 'fb', 'sty', 'drv', 'rec', 'ffin', 'ghw', 'ca', 'gar', 'reg', 'log_lot_rec'].

After applying the general-to-specific approach, the final model only keeps the interaction with recreational room. The final model is now:

Cons ->	b = 7.591	SE = 0.227	t-statistic = 33.505	p-value = 0.0
LogLot ->	b = 0.32	SE = 0.028	t-statistic = 11.562	p-value = 0.0
Bdms ->	b = 0.038	SE = 0.014	t-statistic = 2.68	p-value = 0.008
Fb ->	b = 0.163	SE = 0.02	t-statistic = 8.043	p-value = 0.0
Sty ->	b = 0.091	SE = 0.013	t-statistic = 7.22	p-value = 0.0
Drv ->	b = 0.113	SE = 0.028	t-statistic = 4.018	p-value = 0.0
Rec ->	b = 1.443	SE = 0.626	t-statistic = 2.304	p-value = 0.022
Ffin ->	b = 0.104	SE = 0.022	t-statistic = 4.835	p-value = 0.0
Ghw ->	b = 0.184	SE = 0.044	t-statistic = 4.208	p-value = 0.0
Ca ->	b = 0.166	SE = 0.021	t-statistic = 7.804	p-value = 0.0
Gar ->	b = 0.048	SE = 0.011	t-statistic = 4.206	p-value = 0.0
Reg ->	b = 0.134	SE = 0.023	t-statistic = 5.917	p-value = 0.0
Loglot_Reg ->	b = -0.161	SE = 0.073	t-statistic = -2.213	p-value = 0.027

R-squared = 0.689

g) If the condition of the house is omitted from the regression, the estimated effect of air conditioning is likely to be upward biased. This can be argued as follows:

Better maintained houses should have higher sell prices, so condition, which is omitted, has a positive effect on log price. At the same time, houses with air conditioning are likely to be newer or better maintained, so AC ~~is~~ could be correlated with the condition variable. Therefore, part of the positive effect of house condition on price will be incorrectly attributed to AC, possibly overestimating its causal effect on the sale price.

h) We consider the model with $\log(\text{sell})$ as dependent variable, and $\log(\text{lot})$ along with all other variables in their original form as explanatory (no interaction effects). We estimate the coefficients using the first 400 observations, and then predict $\log(\text{sell})$ for the rest observations. Finally, we comment on the predictive power of the model relative to the variability in $\log(\text{sell})$ of the evaluation sample.

Model parameters after estimation using the first 400 observations:

Cons ->	b = 7.673	SE = 0.292	t-statistic = 26.241	p-value = 0.0
LogLot ->	b = 0.314	SE = 0.036	t-statistic = 8.68	p-value = 0.0
Bdms ->	b = 0.038	SE = 0.017	t-statistic = 2.172	p-value = 0.03
Fb ->	b = 0.152	SE = 0.025	t-statistic = 6.17	p-value = 0.0
Sty ->	b = 0.088	SE = 0.018	t-statistic = 4.85	p-value = 0.0
Drv ->	b = 0.086	SE = 0.031	t-statistic = 2.751	p-value = 0.006
Rec ->	b = 0.055	SE = 0.034	t-statistic = 1.611	p-value = 0.108
Ffin ->	b = 0.115	SE = 0.027	t-statistic = 4.291	p-value = 0.0
Ghw ->	b = 0.199	SE = 0.053	t-statistic = 3.748	p-value = 0.0
Ca ->	b = 0.178	SE = 0.027	t-statistic = 6.521	p-value = 0.0
Gar ->	b = 0.053	SE = 0.015	t-statistic = 3.583	p-value = 0.0
Reg ->	b = 0.151	SE = 0.042	t-statistic = 3.586	p-value = 0.0

R-squared = 0.670

The model's MAE in the evaluation sample is about 0.128 log points, which is roughly interpretable as an average error of about 12.8% in house prices.

The standard deviation of the log of the price in the prediction sample is 0.289

So the MAE relative to the variability in the log of the price is $0.128 / 0.289 = 44.3\%$.

Therefore, the model has reasonably good predictive power, because its average prediction error is substantially smaller than the natural variation of the log price.
